Ayush Ravi Chandran

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Education

University of Massachusetts Amherst — Expected Graduation: May 2026

Bachelor of Science in Computer Science, Bachelor of Science in Mathematics — GPA: 3.85/4.00

- **Coursework**: Algorithms, Networks, Distributed Systems, Graduate Machine Learning and Artificial Intelligence
- **Awards**: Chancellor's Scholarship (40% Tuition), Dean's International Scholarship, Flynn Research Scholarship, Dean's List, Outstanding TA Award
- **Teaching Assistant**: Led discussions, held office hours, and mentored teams through system design for 200+ students

Experience

Software Engineering Intern — September 2025 - Present

Center for Data Science and AI - University of Massachusetts Amherst — *Amherst, MA*

- Built data pipelines (**PostgreSQL**, **S3**) to ingest and normalize multi-organization salt marsh data for a **Massachusetts DEP**-funded initiative, with schema mapping and per-record error reporting
- Eliminated paper-based field collection and manual CSV transcription for **100+** researchers across **20+** organizations, providing live query, edit, and map visualization via a real-time web portal
- Developed a multi-tenant web portal (**React**, **FastAPI**) with role-based access control and containerized deployment to manage **1,000+** weekly field records
- Built a **Flutter** mobile app replacing **KoboToolbox**, enabling nested multi-plot/multi-species survey workflows with **GPS** capture, **RTK** location matching, and a **QC** validation pipeline

Software Engineering Intern — May 2025 - August 2025

Massachusetts Green High Performance Computing Center (MGHPCC) - Unity HPC Cluster — *Holyoke, MA*

- Engineered low-latency telemetry pipelines in a **Linux**-based **HPC** environment to process **12+** months of cluster activity, enabling real-time monitoring and faster incident response
- Optimized **SQL** queries and **data indexing** strategies to accelerate SLURM log analysis by 80%, enabling faster anomaly detection across **10.9M+** job records
- Applied **unsupervised clustering** (DBSCAN) to identify 63,000+ underutilized NVIDIA A100 GPU hours, uncovering **$120K+** in wasted compute resources
- Built **CI/CD** pipelines with **GitHub Actions** and **pytest** to automate linting, formatting, and test coverage checks on a shared Python monitoring module, reducing manual review overhead

Engineering Lead — June 2021 - May 2025

Instilt Educate (Nonprofit) | edu.instilt.com — *Remote*

- Led an engineering team of **10** through sprint planning and backend redesign, building a tutor availability and class matching system (**Express.js**, **PostgreSQL**) that eliminated scheduling conflicts and saved **10+** hours per week
- Developed automation workflows integrating the website, Notion database, email, and Slack to automate volunteer onboarding, interview scheduling, and status notifications for a **300+** member organization

Projects

Poker Bot | *Python, PyTorch, NumPy* | poker-gui.vercel.app — April 2025

- Built a **Deep Q-Network** reinforcement learning agent in **PyTorch** to play Heads-Up Limit Texas Hold'em
- Designed probabilistic game-state encoding using Monte Carlo win-rate estimates, improving action selection
- Trained agent via **self-play** with experience replay, achieving **4th/23** ranking and demonstrating emergent strategies like optimal folding

RouteAble | *TypeScript, NestJS, PostgreSQL, React Native, PyTorch* | github.com/RouteAble — November 2023

- Launched a full-stack app in **36 hours** to crowdsource images of inaccessible locations, enhancing navigation for users with mobility challenges
- Developed a **PostgreSQL** backend with Dockerized **ML models** for obstacle detection and similarity, achieving **92%** accuracy in identifying accessibility barriers
- Received **Most Impactful** Award and a $2,000 prize at the UChicago Winter Tech Showcase '24 for real-world accessibility impact
- Won **Best Use of GitHub** at HackUMass for novel full-stack architecture, awarded by **Major League Hacking**

Skills

Languages: Python, JavaScript, TypeScript, SQL, C, C++

Frameworks: React, FastAPI, Flutter, PyTorch, TensorFlow, Express.js, NestJS, Pandas, scikit-learn

Developer Tools: Git, Docker, Kubernetes, AWS (S3, Lambda), PostgreSQL, Linux, Redis, SLURM, Grafana, pytest